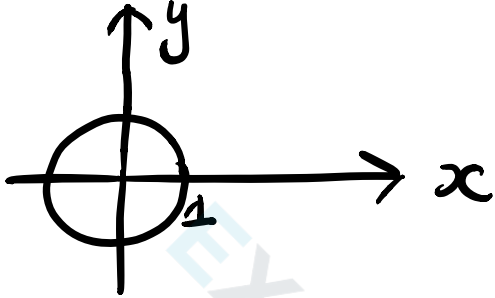


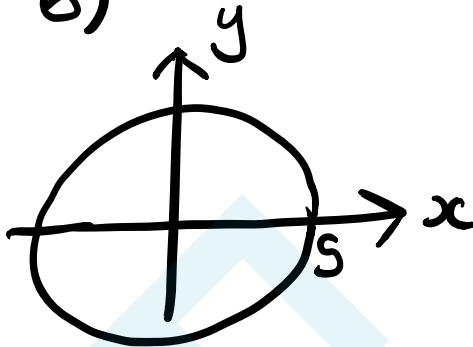
Sketching polar curves

Basic skills

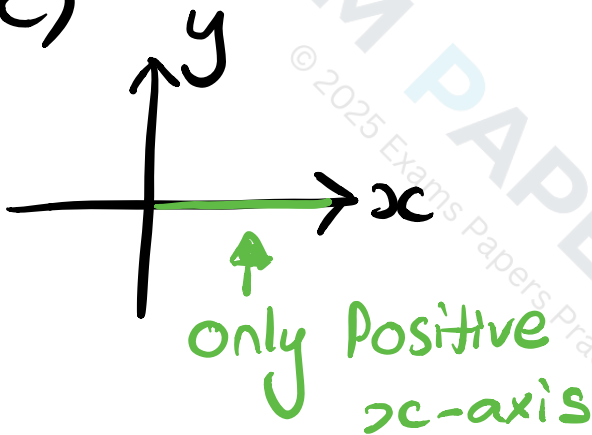
Q1. a)



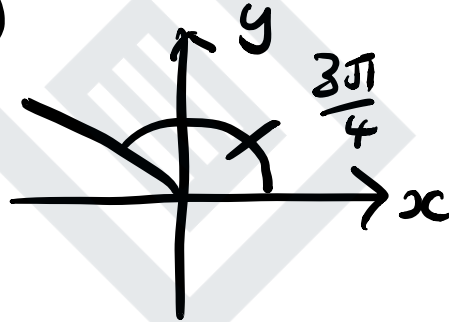
b)



c)



d)



Q2. $r = 2 \sin(3\theta)$

a) $\theta = 0 \Rightarrow r = 2 \sin 0 = 0$

b) $\theta = \frac{\pi}{6} \Rightarrow r = 2 \sin\left(3 \times \frac{\pi}{6}\right) = 2 \sin \frac{\pi}{2} = 2$

c) $\theta = -\frac{\pi}{4} \Rightarrow r = 2 \sin\left(3 \times -\frac{\pi}{4}\right) = -2 \sin \frac{3\pi}{4}$
 $= -\sqrt{2}$

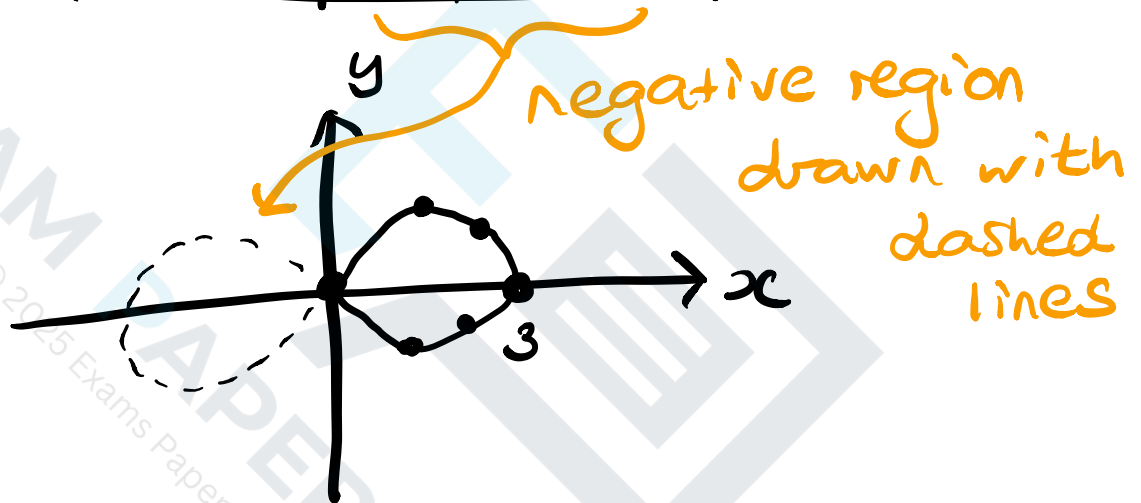
Competency skills

Q3. $r = 3\cos\theta$, $0 \leq \theta < 2\pi$

a)

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$
r	3	$\frac{3\sqrt{3}}{2}$	$\frac{3}{2}$	0	-3	0	$\frac{3}{2}$	$\frac{3\sqrt{3}}{2}$

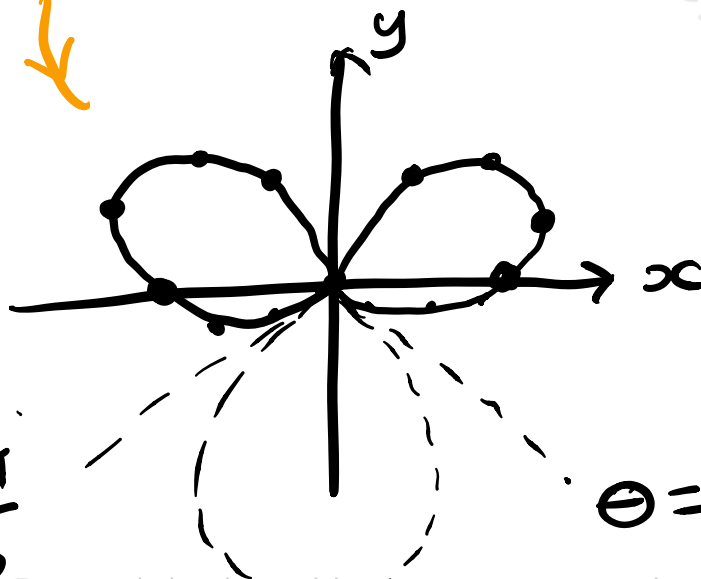
b)



c) symmetric about x -axis, as
 $\cos(2\pi - \theta) = \cos\theta$

Q4. $r = \cos(2\theta) + \sin\theta$, $0 \leq \theta < 2\pi$

Try plotting points



If you struggle
do increments of
 $\frac{\pi}{6}$

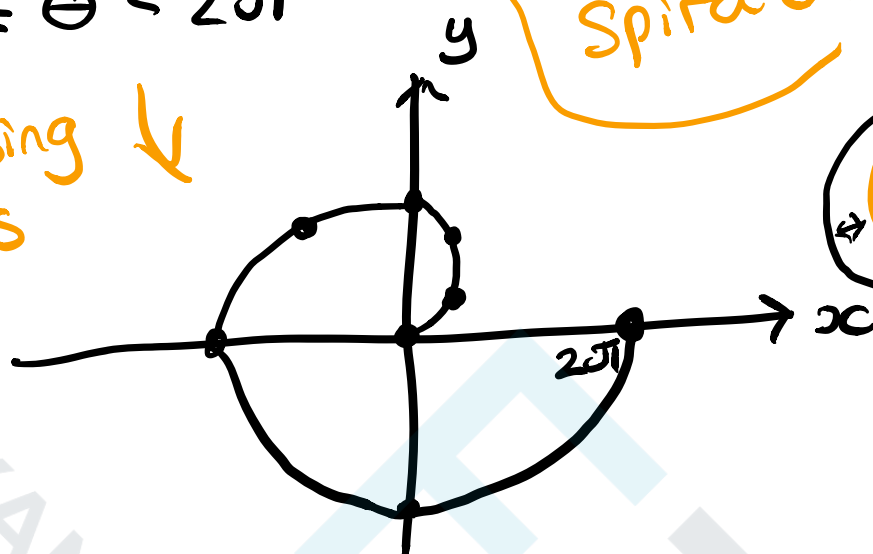
$\theta = \frac{7\pi}{6}$

$\theta = \frac{11\pi}{6}$

Q5. $r = \sin \theta + \theta$

a) $0 \leq \theta < 2\pi$

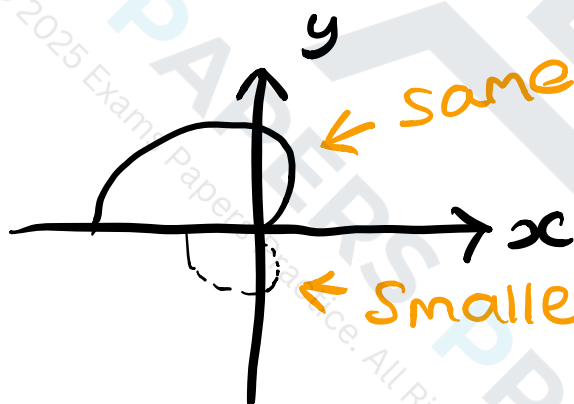
Plot using points



Spiral + oscillations



b) $-\pi < \theta < \pi$



same as above

smaller condensed spiral
but negative so dashed

Advanced skills

Q6. $r = 2 - 4 \sin \theta$

a) need to find $r < 0$, so look at

$$r = 0 \Rightarrow 2 - 4 \sin \theta = 0 \Rightarrow \sin \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$$

all solutions
sin symmetry

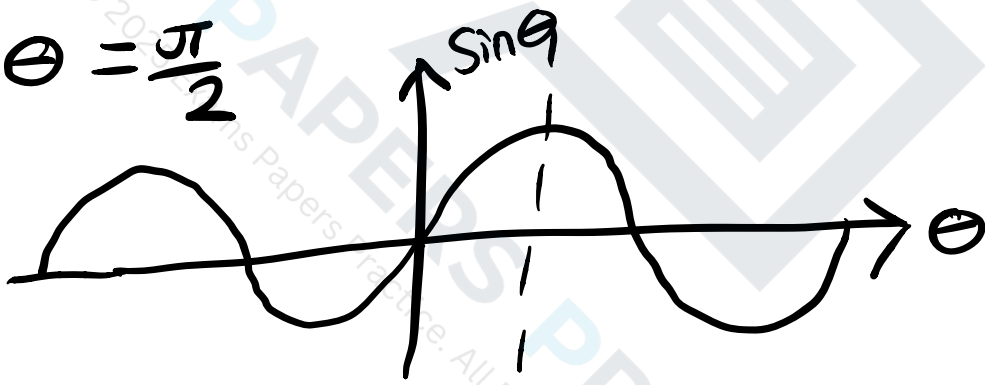
so as $\sin \theta < \frac{1}{2}$ for $\theta < \frac{\pi}{3}$
and $\sin \theta < \frac{1}{2}$ for $\theta > \frac{2\pi}{3}$

we cannot plot

$$\theta < \frac{\pi}{3} \text{ and } \theta > \frac{2\pi}{3}$$

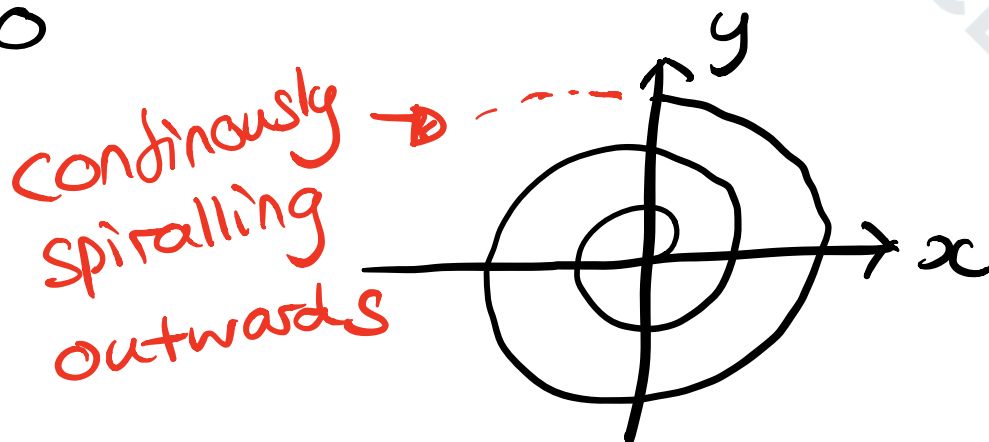
b) Symmetric about y-axis

as $\sin \theta$ has a 'line of symmetry'
at $\theta = \frac{\pi}{2}$



Q7. $r = \theta$ is a spiral as $r \propto \theta$

so



Q8. $r = \tan \theta$, $-\pi < \theta \leq \pi$

a) For $r < 0$ solve $r = 0$

$$\Rightarrow 0 = \tan \theta \text{ so } \theta = 0, \pi, -\pi$$

So as $\tan \theta < 0$ for $-\frac{\pi}{2} < \theta < 0$

and $\frac{\pi}{2} < \theta < \pi$ we cannot plot these regions

Other issues with undefined values

As $\theta \rightarrow \frac{\pi}{2}$ $\tan \theta \rightarrow \infty$ so $r \rightarrow \infty$

As $\theta \rightarrow -\frac{\pi}{2}$ $\tan \theta \rightarrow -\infty$ so $r \rightarrow -\infty$

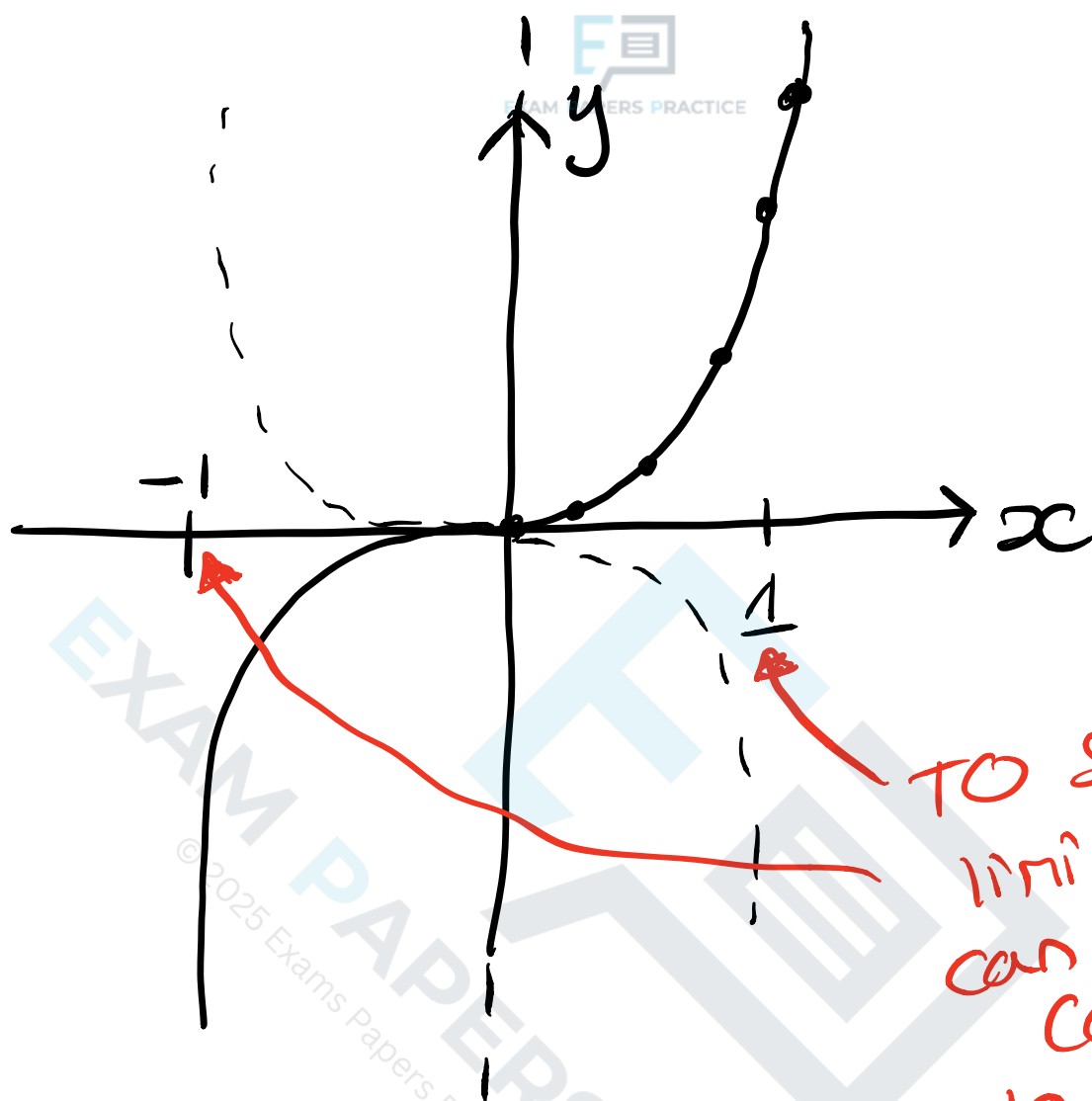
Note that $\theta = \frac{\pi}{2}, -\frac{\pi}{2}$ are undefined

So will not draw

$-\frac{\pi}{2} < \theta < 0$, $\frac{\pi}{2} < \theta < \pi$ and

$$\theta = \frac{\pi}{2}, \theta = -\frac{\pi}{2}$$

b)



TO see this
limit we
can
convert
to cartesian

converting
gives

$$r = \tan \theta$$

$$r \cos \theta = \sin \theta$$

$$x = \frac{y}{r}$$

$$\text{So } x = \frac{y}{\sqrt{x^2 + y^2}}$$

$$x^2 (x^2 + y^2) = y^2$$

$$x^4 + x^2 y^2 = y^2$$

$$y^2(1 - x^2) = x^4$$

$$y = \sqrt{\frac{x^4}{1 - x^2}} = \pm \frac{x^2}{\sqrt{1 - x^2}}$$

As $x \rightarrow \pm 1$ $y \rightarrow \pm \infty$

